

Element J

Jake Steinberg, Astrid Santiago, Caleb Bridgeman, Anne-Marie Georgi, Jack Hanson

Initial email:

Dr. Brossman (Drexel University)

I am an assistant professor at Drexel University teaching cardiovascular and pulmonary physical therapy. In this course we have 60-95 students annually. I teach about objective measures for respiratory evaluation. One objective measure we have is maximal inspiratory pressures or MIP. MIP is a measure of the strength of muscles for taking a breath in -or inhaling. The primary muscle of inhalation at rest is the diaphragm, and other muscles support the diaphragm when someone is maximally exercising or in a diseased state (sternocleidomastoid, trapezius ect).

MIP is of interest to medical providers as it can identify lung diseases that limit inhalation, ventilatory failure and respiratory muscle strength or respiratory muscle insufficiency.

To measure MIP a person generally inhales through a device, a manometer, which records the maximal amount of pressure that a person can generate. Normative values for MIP vary greatly in studies. We generally use MIP with the person as their own baseline and demonstrate changes over time when providing a strengthening routine to the diaphragm with an inspiratory muscle trainer or IMT. I generally like the [Phillips threshold IMT](#). An IMT is a device that a person breathes through which has a pressure load that you can increase or decrease like weight training for the diaphragm. You measure IMT after a period of training and compared to a person's baseline MIP to see if you have changed strength over time.

We have a cohort of 97 this year, and have 1 (one) powerbreathe K series device to measure MIP ([POWERbreathe K-Series IMT](#)) and it was \$2,000 and it is not economical to purchase enough devices for this class. I also only have a few Phillips Threshold IMT and they are \$100.00 and neither are covered by insurance. I want to allow ALL my students to measure their own MIP in class, and to experience IMT as we know this translates to use of this measurement for their patients as they move to become excellent physical therapists.

The limiters/barriers in terms of cost for these devices often impact those who need them the most as they are quite expensive. During COVID many more underrepresented minorities died due to respiratory dysfunction. I believe the algorithms used to measure respiratory impairment were flawed or not used. Some of these when not measured are calculated with flawed metrics based on race. I want students to measure directly and not use a flawed calculation. I purchase a negative inspiratory force (NIF) meter ([Negative Inspiratory Force \(NIF\) Meters](#)) which is also a few hundred dollars.

What I need: a safe (able to be cleaned and shared between students), inexpensive, reliable and valid way to measure MIP with enough devices to be able to support a large cohort in minimal time. For our cohort of 97 students the lab is an hour during which they rotate into a MIP measuring station but we need enough devices for a number of students (around 10-20) to use at one.

Documentation:



STEINBERG, JACOB <jacob.steinberg@student.colonialsd.org>
to hb456 ▾

Thu, Jan 23, 10:42 AM ☆ ↶ ⋮

Dear Dr. Brossman,

I hope you're doing well. I wanted to share an update on our progress. We have completed a prototype, and I've attached a picture and a video for your reference. We will add labels to indicate the challenge level and pressure readings as we proceed.

Please let me know if you have any questions or feedback. I appreciate your time and look forward to your thoughts.

Have a great day,

Jake



Brossman, Heather ✓
to me ▾

Jacob,

I am so interested in seeing this in person.

I may have to schedule time to come see your class, and discuss your progress.

This is awesome!

Dr. Brossman



STEINBERG, JACOB <jacob.steinberg@student.colonialsd.org>
to Heather ▾

Jan 27, 2025, 11:44 AM ☆ ↶ ⋮

Dear Dr. Brossman,

I'm glad to hear that you're interested. For the next few weeks, our classes follow a regular schedule from 11:33 AM to 12:02 PM and 12:41 PM to 1:12 PM. Since our break is designated for lunch, it would be more convenient to meet during one of these half-hour periods.

However, I believe the best approach would be to coordinate the meeting through Mr. Muscarella. Please let me know how you'd like to proceed.

Best regards,
Jake

Jim,
I wanted to let you know we have a budget for the students working on the maximal inspiratory pressure devices.
I have cc'd Jay here as he will handle the ordering of materials from the budget.

We have \$80 for each group working on the MIP. The students can ask him for the materials they need and he will work out the details for getting the materials, and ordering.

We would like the students to keep track of their budget and spending to share at the end of their work on this project. The students sent me emails asking for some clarifying questions about the project which I answered. If you would like me to directly let them know about the budget I am happy to do so- otherwise I will have you let them know.

I look forward to hearing how the project is moving forward, and am so pleased that Drexel University was excited to support this endeavour.

Let me know if there is anything I can do to support.

Sincerely,
Heather

Dear Dr. Brossman,

I hope this message finds you well. My name is Jake Steinberg, and I am the liaison for one of the groups in Mr. Muscarella's engineering class. We are currently in the early stages of our design process and want to ensure that our approach aligns with your expectations.

Based on your previous email and our recent conversation, we have compiled a table of design requirements, which I've included below for your review. As part of our process, we need feedback from stakeholders—including yourself and potentially some of your students—on these requirements. Specifically, we'd appreciate your thoughts on prioritization and any additional suggestions you might have.

Additionally, I want to let you know that the other group in the class will be reaching out soon, so please keep an eye out for their email as well.

Hello Jake,
This is an excellent and precise synopsis of my presentation and "ask".
I agree that the top priorities are safety, repeatability, and accuracy.
I am less concerned with portability as it can live in our labs for student use with a prototype, and size could be handled if needed by countertop space or a traveling carts if it is larger than a hand held device.

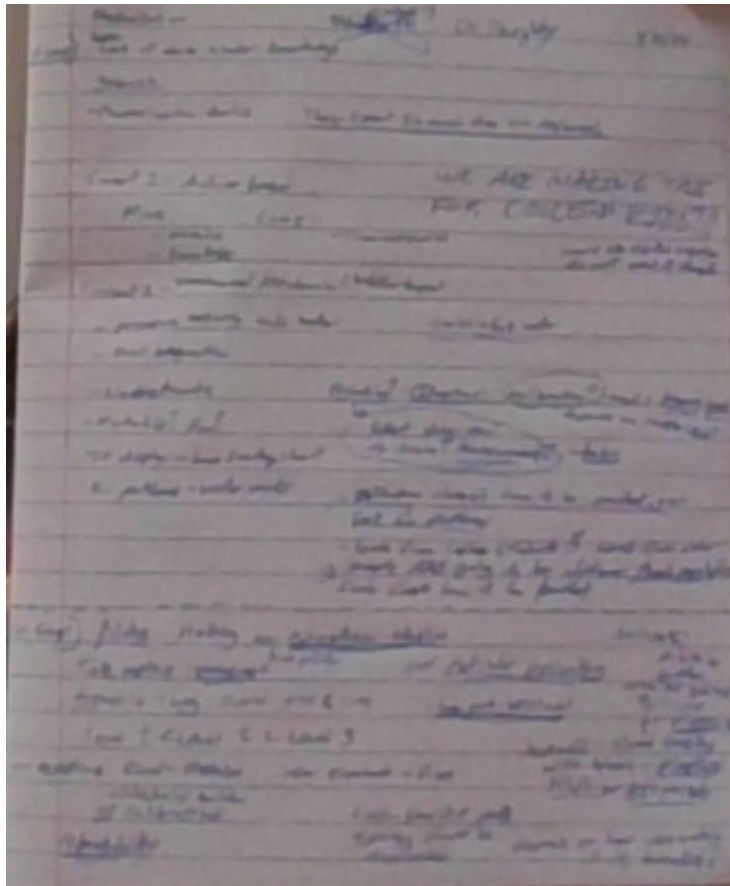
I am still working on some funding if I can get it, but as of right now I don't have a budget to give you.
I will continue to work on this.

Thank you for this communication! I look forward to hearing of your progress.
Dr. Brossman

Involvement of stakeholders and field experts	Dr. Brossman (Stakeholder)	Dr. Brossman's students (Stakeholder)	Any future patients (Stakeholder)	Mr. Scheidley (Expert)	Mr. Flannagan (Expert)	Dr. Sarver (Expert)
Comprehensive	As we are	Dr. Brossman	Along the	Jake went to	Jack, from our	The Zoom

evaluation	working with Doctor Brossman to build a device for her class, her feedback was very important to the success of our group's project. While presenting our designs to Dr. Brossman, she gave us great feedback on the importance of each priority, and how we can achieve our goals. She also informed us on the basics of what an MIP	wants to make sure her students have a proper appreciation for being able to test a patient's respiratory health. Without an appropriate appreciation for it, they could brush past it in their future practices, which could lead to problems with patients down the line.	same line, any future patients being treated by Dr. Brossman's students would want them to be familiar with the procedures to test someone's lung capacity.	see Mr. Scheidley twice over the course of the project. The first was to help figure out a general solution to the problem, which we used as a possible design. The second time was to help sort out the issue of controlling the water flow into the bottom of the tube. He helped create	group, showed the device to Mr. Flanagan. Jack explained that it was difficult to make calculations. He suggested that we would use slits because slits have a discrete number to them.	call with Dr. Sarver was very beneficial. She mentioned that this could be a good educational device. If the purpose of our project was to show how the smaller the hole on the pvc pipe, the more resistance there is and the harder it is to bring the water up.
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	device is			the idea of using slits, which is what we're going to do in our second iteration.		
Synthesizing the project external evaluations						



- Based on our research, we have multiple questions that will help enhance our knowledge of the Drexel project. The questions include: Do we need some sort of screen/device to show real time measurements? Will it be connectable to some sort of program? Do we need to code this? Will it work with software for data analysis also what formats will it support? Will the device be digital or analog, and what are the pros and cons of the chosen format? What is the intended user experience for students? Should the device be easy to use independently, or will there be assistance from instructors during measurement? Are there specific materials wanted to ensure easy and deep cleaning? What are the specific sanitation protocols? Are they giving us the funding to build this? Is the goal to improve an existing MIP device or create a completely new one?

Questions and answers:

NOTES (Dr. Heather Brossman)

NOTES

- Weight training for the respiratory muscles analogy - how to know when the limits to achieve the purpose of exercise (ex: strengthening muscles)
- Diaphragm
 - Has a dome shape, helping contract and expand the lungs
 - Change of pressure helps the air go from the atmospheric pressure to bodily pressure
 - Two muscles together make it
- To help ventilate:
 - Make diaphragm stronger
- Weaken the organ:
 - Injuries from surgeries
 - Car crashed
 - Kicked
 - Obesity
 - Cachexia
 - Neuromuscular disease like ALS, multiple sclerosis
 - COVID-19
- MIP
 - Brings the air in during therapy
 - Tears the tube by taking everything out

- Has the 1500 power device, but that might be another problem (she needs more of them)
 - Hard to clean
- MAP
 - Bring the air out
- Strengthening Inspiratory Muscles
 - What we have
 - Super expensive

PROBLEM LIST

- Need to teach respiratory measurement for clinicians
- Need reasonable cost measurement device
- Need ease of use for cleaning
- A lack of accessible devices

QUESTIONS

- Would it be easier to have cleaning stations (15 devices per station) or easier to have detachable mouth pieces?
 - Combination of the two, but she really wants our opinion on it since it is a lot of pieces.
- How precise does it need to be (to the nth degree)?
 - She not doing research about it, so she wants to know that it is reliable enough so that, with multiple trials with similar-ish numbers the outcomes are the same
 - Precision over accuracy
- Does the needle need to stay where it is so that people can read? Screen better?

- Definitely has to stay because it is very hard to miss. Stay around the peak number.
- How portable?
 - Doesn't have to be super portable
- Want the scale to reach 200 (max pressure)
- How many years is this going to be used for?
 - Love this to last for 15 years
 - Shouldn't have too many drops, but make sure that it is durable enough
 - Some level of protection
- What is the intended user experience for students? Should the device be easy to use independently, or will there be assistance from instructors during measurement?
 - Individually make it more interesting for everyone
 - Cleaning it is becoming a problem since there is a large number of people moving around
- What are the specific sanitation protocols?
 - Nothing yet
- Are they giving us the funding to build this?
 - Pending
- Is the goal to improve an existing MIP device or create a completely new one?
 - A HUGE Improvement or create a new one
- Are there specific materials wanted to ensure easy and deep cleaning?
 - Something that can withstand a limited amount of water
- Will it work with software for data analysis also what formats will it support?

- Up to us
- Will the device be digital or analog, and what are the pros and cons of the chosen format?
 - Up to us